MRISystemPhantom Enables Reinforcement-Learning-Based Adaptive T_1 Mapping

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Introduction

Quantitative MRI protocols conventionally fix their acquisition parameters before scanning so that they remain informative across a wide range of unknown tissue properties. This robustness can spend limited scan time on redundant measurements. Adaptive MRI instead uses measurements already acquired to select the next acquisition, but developing such policies directly on scanner hardware is costly and provides no exact tissue-parameter ground truth [2]. We developed *MRISystemPhantom.jl*, an executable digital twin of the ISMRM/NIST Model 130 system phantom [3], and used it to test whether reinforcement learning (RL) can reduce simulated T_1-estimation error relative to fixed schedules.

Methods and materials

MRISystemPhantom.jl implements the phantom's water-filled housing, T_1, T_2 and proton-density plate geometry, fiducial grid, field-strength dependent T_1/T_2 reference values and episode-level tissue randomisation. It returns standard KomaMRI objects [1] and supplies the sequence generation, reconstruction, region-of-interest extraction and signal fitting required for an end-to-end quantitative simulation.

Each episode acquired noisy two-dimensional inversion-recovery spin-echo images of five spheres whose T_1 values were independently resampled over 0.04–1.9 s. Complex Gaussian k-space noise produced a mean NEMA MS-1 signal-to-noise ratio of approximately 17 at the reference acquisition. After each image, the current T_1 values were refitted and a proximal-policy-optimisation agent [4] chose the next inversion and repetition times under a 560 s maximum scan budget. The best policy additionally observed a 12-bin histogram of the inversion times already executed. Training progressed from an analytic signal model through cached-water simulations to full Bloch simulation. Frozen policies were tested over 24 held-out episodes against a policy without acquisition history and two fixed schedules. The primary endpoint was mean absolute percentage error (MAPE) in the fitted T_1 values, with bootstrap 95% confidence intervals.

Results

The TI-history policy achieved a mean T_1 MAPE of **2.93%** (95% CI: 2.58–3.30%), compared with **6.04%** (5.36–6.70%) for a fixed log-grid schedule under the same maximum budget: a 51.4% relative reduction. Its 90th-percentile error was 4.19%, and 95.8% of episodes finished below 5% error, compared with 7.81% and 25.0%, respectively, for the fixed schedule. Removing the executed-TI histogram increased mean error to 4.16% (3.41–4.99%). The history-aware policy remained below 3.2% mean error in the short, intermediate and long realised-T_1 bands.

Discussion

Explicit acquisition history improved the policy despite a similar number of measurements, indicating that the running T_1 estimates alone did not fully describe which parts of inversion-time space had already been sampled. The digital phantom made this comparison reproducible across known, continuously varying tissue properties. The positive result is limited to a five-sphere, simulator-only proof of concept with one training seed and 24 evaluation episodes; the policy did not outperform fixed coverage on the complete 14-sphere plate.

Conclusion

MRISystemPhantom.jl provides a reusable quantitative testbed for adaptive MRI development. Within a bounded simulated T_1-mapping task, an RL policy that remembered its previous inversion times substantially outperformed fixed acquisition schedules.

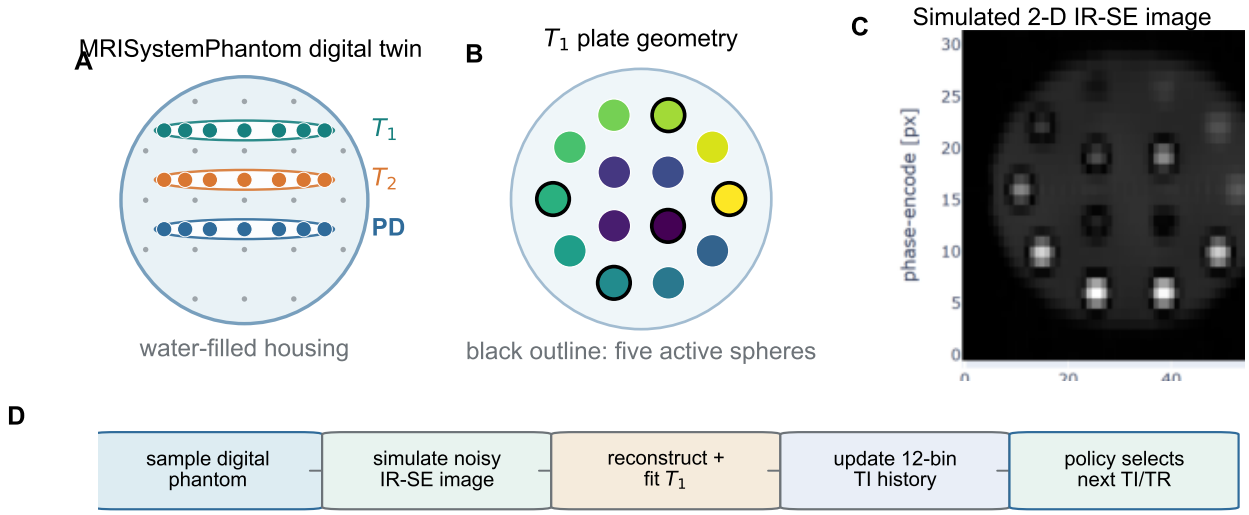


Figure 1: MRISystemPhantom and the adaptive acquisition method. A: Configurable digital representation of the Model 130 phantom. **B:** T_1-plate geometry; black outlines identify the five active positions whose values are resampled each episode. **C:** Example Bloch-simulated IR-SE reconstruction. **D:** Closed-loop experiment in which the policy uses fitted T_1 values and an executed-TI histogram to select the next timing.

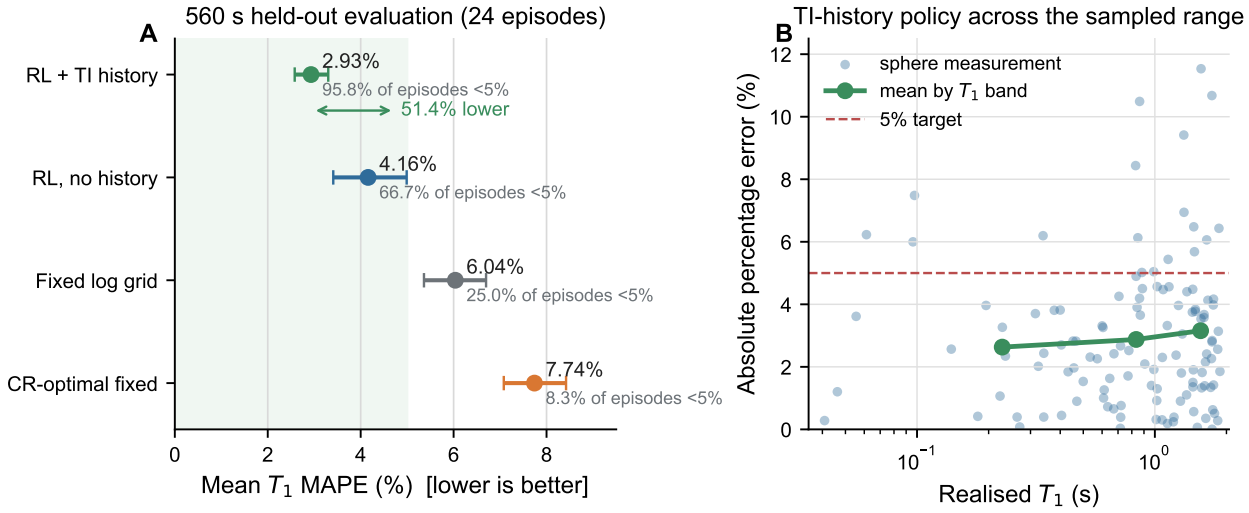


Figure 2: Five-sphere held-out result. A: Mean fitted-T_1 error with 95% confidence intervals; percentages beneath each point give the fraction of episodes below 5% error. **B:** Absolute error for all 120 held-out sphere measurements, with means over short, intermediate and long realised-T_1 bands.

References

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Acknowledgements

This work formed part of Arthur Allilaire's BEng final-year project at Imperial College London.